

OSH Universal Substrate Benchmark

Cross-ISA Compilation, RISC-V Virtual Golden Parity, and
Physical ESP32-S3 Validation

21 / 22	2 / 2	$\leq 5e-8$
CROSS-TARGET COMPILE PASS	RV32IMF GOLDEN CASES	MAX VIRTUAL GOLDEN ERROR

Evidence scope. Cross-target object generation, execution of the original hash-verified Tier-A harness under an emulated RISC-V32 environment, and separately documented physical ESP32-S3 validation.

Validation date: 23 September 2026

What was demonstrated

Evidence	Measured result	Status
Artifact integrity	Core + header + original Tier-A harness SHA-256 matched	PASS
Cross-target compilation	21 of 22 target profiles produced objects	PASS
RV32IMF virtual execution	Original Tier-A harness executed under QEMU	PASS
OSH_PRESENCE	Deform 5e-8; final 5e-8	PASS
OSH_NO_ACTIVATION	Deform 2e-8; final 2e-8	PASS
Original harness verdict	OVERALL=PASS	PASS
Physical ESP32-S3	Two-device Tier-A reproduction + scaling + Tier-B1	PASS

The central result is a portability chain: the frozen Tier-A substrate cross-compiles broadly, the original golden-validation harness reproduces its reference behavior on an emulated RV32IMF platform, and physical ESP32-S3 evidence remains a distinct physical layer.

Frozen reference, variable platform

Artifact	SHA-256
osh_homeostatic_core.c	e09f1d7928f291a64ab5c8a8d6f9af62cb2b293a1022e6897044fd04288fd363
osh_homeostatic_core.h	21b4caf8308044b91c20bcbf43d615289e5826f1a8aeb4cef90a7c575082a25
osh_tier_a_main.c	ef96ab1fc52df2cd057115cff7fd4e253dbe614be2c50b10404e551dd59c679d
RV32IMF QEMU ELF	1e69e673b8776af92abde963610631764369364e1b915884b580fe7cf12ff5d1

The decisive RISC-V run reported **CORE_CHANGED=false, HEADER_CHANGED=false, HARNESS_CHANGED=false**. Platform compatibility code supplied only the environment required by the original ESP32-oriented harness.

Claim boundary: compilation is not execution; emulation is not physical silicon; adapter timing and adapter heap/PSRAM values are not treated as performance measurements.

Cross-target compilation pressure test

Target	Compile	Object bytes
x86_64	PASS	4,072
i386	PASS	3,316
AArch64	PASS	3,896
Cortex-M0	PASS	3,696
Cortex-M3	PASS	3,788
Cortex-M4	PASS	2,704
Cortex-M7	PASS	2,628
Cortex-M33	PASS	2,636
RV32I	PASS	7,820
RV32IM	PASS	7,664
RV32IMF	PASS	6,292
RV32IMC	PASS	6,852
RV64GC	PASS	8,240
MIPS32EL	PASS	4,684
MIPS64EL	PASS	5,800
PowerPC32	PASS	4,048
PowerPC64	PASS	5,544
SPARC	PASS	3,816
SPARC V9	PASS	5,544
m68k	FAIL	0
LoongArch64	PASS	4,040
WASM32	PASS	2,249

Result: 21/22 PASS. The m68k failure is retained. This phase demonstrates compilation portability only.

RISC-V32 virtual execution

RV32IMF ISA PROFILE	ilp32f ABI	QEMU 11.1 VIRTUAL MACHINE	PASS EXECUTION
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The original Tier-A harness was compiled and executed under a RISC-V32 virtual machine. Floating-point state was enabled before entering C, matching the requirements of code compiled for the RISC-V F extension.

The final run reached both original golden cases and returned the harness's own PASS verdicts.

OSH_PRESENCE

State	RV32IMF	Golden
Deformed C	0.92000001	0.92000001
Deformed P	0.84948885	0.84948885
Deformed X	0.00000000	0.00000000
Deformed T	0.36296293	0.36296293
Deformed H	0.89913380	0.89913374
Final C	0.87287801	0.87287801
Final P	0.75194156	0.75194156
Final X	0.02946170	0.02946170
Final T	0.35200831	0.35200828
Final H	0.83726954	0.83726948

5e-8 DEFORM MAX ABS ERROR	5e-8 FINAL MAX ABS ERROR	PASS ORIGINAL VERDICT
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Top-K: pass1 = 1 -> 2 -> 3; pass2 = 1 -> 2 -> 3.

OSH_NO_ACTIVATION

State	RV32IMF	Golden
Deformed C	0.92000001	0.92000001
Deformed P	0.81615549	0.81615549
Deformed X	0.00000000	0.00000000
Deformed T	0.39296296	0.39296293
Deformed H	0.84913378	0.84913378
Final C	0.88257604	0.88257604
Final P	0.74572491	0.74572491
Final X	0.06080713	0.06080714
Final T	0.40105211	0.40105208
Final H	0.79903817	0.79903817

2e-8 DEFORM MAX ABS ERROR	2e-8 FINAL MAX ABS ERROR	PASS ORIGINAL VERDICT
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Top-K: pass1 = 4 -> 5 -> 3; pass2 = 4 -> 5 -> 3.

Golden parity

Case	Deform error	Final error	Threshold	Verdict
OSH_PRESENCE	5e-8	5e-8	$\leq 1e-6$	PASS
OSH_NO_ACTIVATION	2e-8	2e-8	$\leq 1e-6$	PASS

The virtual errors are approximately 20x below the acceptance limit for OSH_PRESENCE and 50x below it for OSH_NO_ACTIVATION. These ratios describe only the two frozen cases tested.

The original harness concluded **OVERALL=PASS**; the runner independently recorded **ORIGINAL_TIER_A_QEMU_GOLDEN_PARITY=PASS**.

Independent ESP32-S3 evidence

Evidence item	Measured result
Target	ESP32-S3 rev v0.2; 2 cores; 160 MHz boot log
Memory	32 MB Flash; 16 MB Octal PSRAM
Tier-A Unit #1	Golden parity PASS; 10,000-cycle timing PASS
Tier-A Unit #2	Independent cross-device reproduction PASS
Tier-A timing	29.4728 us/cycle Unit #1; 29.4721 us/cycle Unit #2
Scaling	10 / 64 / 256 / 1024 candidates PASS
Tier-B1 HAL	4 validation cases PASS; 64.0735 us/focus
Regression	Tier-A remained PASS after B1 integration

Physical measurements are not inferred from QEMU. The virtual adapter's timer and memory values are compatibility values and are excluded from performance claims.

From source portability to physical execution

Stage	Question	Result
1. Frozen identity	Are the reference artifacts unchanged?	PASS
2. Cross-target compile	Can the core generate objects across diverse targets?	21/22 PASS
3. Virtual execution	Can original Tier-A execute on RV32IMF?	PASS
4. Golden parity	Do both original golden cases remain within threshold?	2/2 PASS
5. Selection	Are observed Top-K routes preserved?	PASS
6. Physical silicon	Does Tier-A reproduce on real ESP32-S3?	PASS
7. Cross-device	Does a second physical unit reproduce Tier-A?	PASS
8. Layering	Does Tier-B1 run while Tier-A regression remains PASS?	PASS

What this benchmark does not prove

Not claimed	Reason
21 architectures executed OSH	Only RV32IMF completed the new virtual golden execution
QEMU timing equals hardware timing	Virtual timing depends on host/adaptor and is excluded
Virtual heap/PSRAM are measurements	They are adapter compatibility values
m68k portability	The tested profile failed compilation
Full production stack migrated	This benchmark is Tier-A focused
Universal numerical identity	Results apply to the frozen cases and acceptance policy tested
Emulation equals hardware	Emulation complements but does not replace physical validation

A reproducible portability chain

Within the tested scope, the frozen OSH Tier-A computational substrate generated target objects for 21 of 22 tested profiles. The original hash-verified Tier-A harness then executed under an emulated RV32IMF environment and reproduced both original golden cases within the pre-existing $1e-6$ numerical threshold.

OSH_PRESENCE reproduced with maximum deform and final errors of $5e-8$. OSH_NO_ACTIVATION reproduced with maximum deform and final errors of $2e-8$. The observed Top-K candidate order was preserved in both passes of both cases, and the original harness returned OVERALL=PASS.

These virtual results complement a separate physical ESP32-S3 evidence chain: two-device Tier-A reproduction, controlled scaling through 1024 candidates, Tier-B1 HAL validation, and Tier-A regression protection.

Supported statement: the evidence demonstrates broad cross-target compilation portability for the frozen Tier-A core, golden-parity execution on an emulated RISC-V32 platform, and independently documented physical execution on ESP32-S3.

It does not establish execution on every compiled ISA or migration of the complete OSH production stack.

Why the new benchmark is stronger than the compile-only assay

The earlier portability assay established that the frozen OSH substrate could be translated across multiple target families. That result was necessary but incomplete: a compiler can emit target code without proving that the program executes correctly on the target ISA.

The new evidence closes that gap for RISC-V32. The original Tier-A harness - not a replacement harness - was hash-verified, compiled for RV32IMF, executed under QEMU, and allowed to evaluate its own original golden cases.

Evidence question	Earlier assay	Final dossier
Can the frozen core compile for many targets?	Yes	Yes - 21/22 profiles
Is a different ISA actually executing code?	Not established	Yes - RV32IMF/QEMU
Is the original Tier-A harness preserved?	Not required	Yes - SHA-256 locked
Are original golden states compared?	No	Yes - both cases
Does original verdict logic return PASS?	No	Yes - OVERALL=PASS
Is physical silicon kept separate?	Yes	Yes

This distinction is central to the public claim. The new benchmark does not merely add another target name; it adds a higher evidence class.

Evidence chain and reproducibility logic

Stage	Frozen/variable	Evidence
Artifact identity	Frozen	Core/header/harness SHA-256 gates
Compilation targets	Variable	21/22 target object generation
Virtual ISA	Variable platform	RV32IMF / ilp32f / QEMU 11.1
Tier-A cases	Frozen	OSH_PRESENCE; OSH_NO_ACTIVATION
Golden states	Frozen	Original deformed/final references
Acceptance threshold	Frozen	Maximum absolute error <= 1e-6
Virtual result	Measured	2/2 PASS; OVERALL=PASS
Physical result	Independent	ESP32-S3 Tier-A + scaling + Tier-B1

The purpose of this design is to prevent the platform adapter from becoming the experiment. The adapter may supply platform services, but the computational reference and its verdict logic remain fixed.

The final RV32IMF executable is independently anchored by SHA-256:
1e69e673b8776af92abde963610631764369364e1b915884b580fe7cf12ff5d1.

Golden case transcript - OSH_PRESENCE

Field	Virtual result	Golden
Deformed C	0.92000001	0.92000001
Deformed P	0.84948885	0.84948885
Deformed X	0.00000000	0.00000000
Deformed T	0.36296293	0.36296293
Deformed H	0.89913380	0.89913374
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Final X	0.02946170	0.02946170
Final T	0.35200831	0.35200828
Final H	0.83726954	0.83726948

deform_max_abs_error = 0.00000005

final_max_abs_error = 0.00000005

verdict = PASS

Observed Top-K: pass1 = 1 -> 2 -> 3; pass2 = 1 -> 2 -> 3. Candidate ordering therefore remained consistent with the original harness output for this case.

Golden case transcript - OSH_NO_ACTIVATION

Field	Virtual result	Golden
Deformed C	0.92000001	0.92000001
Deformed P	0.81615549	0.81615549
Deformed X	0.00000000	0.00000000
Deformed T	0.39296296	0.39296293
Deformed H	0.84913378	0.84913378
Final C	0.88257604	0.88257604
Final P	0.74572491	0.74572491
Final X	0.06080713	0.06080714
Final T	0.40105211	0.40105208
Final H	0.79903817	0.79903817

deform_max_abs_error = 0.00000002

final_max_abs_error = 0.00000002

verdict = PASS

Observed Top-K: pass1 = 4 -> 5 -> 3; pass2 = 4 -> 5 -> 3.

Public claim language for technical review

A precise public statement supported by the evidence is:

A frozen OSH Tier-A computational substrate generated target objects for 21 of 22 tested cross-target profiles. The original hash-verified Tier-A validation harness was then executed under an emulated RV32IMF environment and reproduced both original golden cases within the pre-existing numerical acceptance threshold, with maximum absolute errors no greater than $5e-8$. These virtual results complement separately documented physical ESP32-S3 validation.

The following stronger statements are intentionally **not** made: that all 21 compiled ISAs have executed OSH; that QEMU timing represents hardware timing; that adapter heap/PSRAM values are physical measurements; or that the entire production OSH stack has been migrated.

Website-ready evidence summary

Card	Website copy
21 / 22 target profiles	Frozen Tier-A core generated objects across x86, ARM, RISC-V, MIPS, PowerPC, SPARC, LoongArch64 and WASM.
RISC-V execution	Original hash-verified Tier-A harness executed under RV32IMF/QEMU.
2 / 2 golden cases	OSH_PRESENCE and OSH_NO_ACTIVATION both returned PASS.
<= 5e-8 max error	Virtual golden parity remained far inside the original 1e-6 acceptance threshold.
Physical ESP32-S3	Independent physical evidence includes two-device Tier-A reproduction, scaling through 1024 candidates and Tier-B1.
Auditability	Frozen artifact hashes and RV32IMF ELF hash are recorded in the dossier.

Recommended button label: **Download Universal Substrate Benchmark.**

Recommended supporting line: **Compile portability -> virtual golden parity -> physical silicon.**

Integrity ledger

Artifact	SHA-256
Frozen core	e09f1d7928f291a64ab5c8a8d6f9af62cb2b293a1022e6897044fd04288fd363
Frozen header	21b4caf8308044b91c20bcfb43d615289e5826f1a8aecb4cef90a7c575082a25
Original Tier-A harness	ef96ab1fc52df2cd057115cff7fd4e253dbe614be2c50b10404e551dd59c679d
RV32IMF final ELF	1e69e673b8776af92abde963610631764369364e1b915884b580fe7cf12ff5d1

These hashes identify the tested artifacts without publishing the complete proprietary implementation.

The decisive execution reported **CORE_CHANGED=false**, **HEADER_CHANGED=false**, and **HARNESS_CHANGED=false**.

Final assessment

The evidence now forms a coherent progression rather than a collection of disconnected tests.

First: broad target compilation shows that the Tier-A substrate is not syntactically tied to one MCU toolchain.

Second: RV32IMF execution demonstrates that the original Tier-A validation program can run on a different ISA and reproduce its frozen numerical references. **Third:** the separate ESP32-S3 dossier provides physical-silicon evidence and cross-device reproduction.

The strongest defensible interpretation is therefore not 'universal hardware support'. It is that the tested OSH Tier-A substrate exhibits **broad compile portability, demonstrated cross-ISA virtual golden parity, and independent physical embedded validation.**

That distinction makes the dossier stronger, not weaker: every claim is tied to the evidence class that actually supports it.

OSH GLOBAL SYSTEMS | Fabián Vallejos | 2026